

Translation in the 21st C.: Human Language Translators + AI

Bruce D. Marron

June 4, 2026

Is AI a threat to human language translators?

The answer is perhaps...but let's investigate!

What is AI afterall?

Artificial Intelligence (AI) and Machine Learning (ML)

Artificial intelligence: the general ability of computers to emulate human thought and perform tasks in real-world environments.

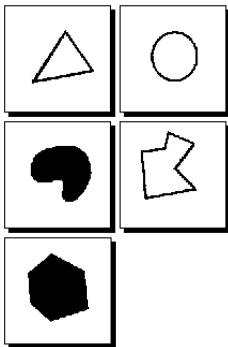
Machine learning: the technologies and algorithms that enable systems to identify and recognize patterns, make decisions, and improve themselves through experience and data.

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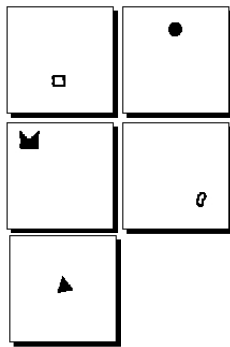
Machine learning is a pathway to artificial intelligence.

Machine Learning (ML)

Training Examples: Class 1



Training Examples: Class 2



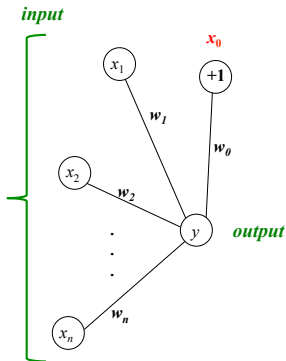
Test example: Class = ?



Perceptrons as simplified “neurons”

w_0 is called the “**bias**”

$-w_0$ is called the “**threshold**”



Input is $(x_1, x_2, \dots, x_n)$

Weights are $(w_1, w_2, \dots, w_n)$

Output y is $+1$ (“the neuron fires”) if the sum of the inputs times the weights is greater or equal to the threshold:

*If $w_1x_1 + w_2x_2 + \dots + w_nx_n > \text{threshold}$
then $y = 1$, else $y = -1$*

*If $w_1x_1 + w_2x_2 + \dots + w_nx_n > -w_0$
then $y = 1$, else $y = -1$*

*If $w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n > 0$
then $y = 1$, else $y = -1$*

*If $w_0x_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n > 0$
then $y = 1$, else $y = -1$*

A Few Definitions

Acronyms in AI Translation

LLM	Large Language Model
NMT	Neural Machine Translation [Google]
GPT	Generative Pre-trained Transformer [OpenAI]
API	Application Programming Interface
SDK	Software Development Kit
IDE	Integrated Development Environment
MIME	Multipurpose Internet Mail Extension; type of file format

Public AI Translators (free with limitations)

Google Translator - Simple Prose

I love going to school! It's such a challenging environment.

↔ español inglés francés ▼

¡Me encanta ir a la escuela! Es un entorno muy desafiante.

Google Translator - Poetry

"For what is it to die but to stand naked in the wind and to melt into the sun? And when the earth shall claim your limbs, then shall you truly dance." (Kahlil Gibran, artist and poet)

→ español inglés francés ▼

"¿Qué es morir sino permanecer desnudo en el viento y fundirse con el sol? Y cuando la tierra reclame tus miembros, entonces podrás bailar de verdad".

The Anthropocene is a proposed new geological epoch based on the observation that human impacts on essential planetary processes have become so profound that they have driven the Earth out of the Holocene epoch in which agriculture, sedentary communities, and eventually, socially and technologically complex human societies developed. The formalization of the Anthropocene as a new geological epoch is being considered by the stratigraphic community, but regardless of the outcome of that process, it is becoming apparent that Anthropocene conditions transgress Holocene conditions in several respects. The knowledge that human activity now rivals geological forces in influencing the trajectory of the Earth System has important implications for both Earth System science and societal decision making. [Steffen et. al. (2018) Trajectories of the Earth System in the Anthropocene, *PNAS*]

Google Translator - Science Text



español

inglés

francès

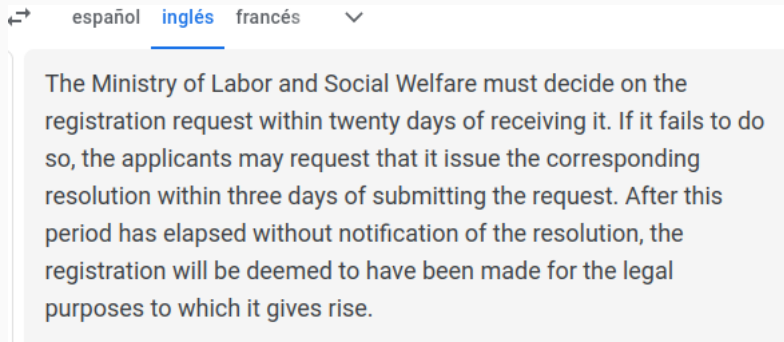


El Antropoceno es una nueva época geológica propuesta que se basa en la observación de que los impactos humanos en los procesos planetarios esenciales se han vuelto tan profundos que han hecho que la Tierra salga de la época del Holoceno, en la que se desarrollaron la agricultura, las comunidades sedentarias y, finalmente, las sociedades humanas social y tecnológicamente complejas. La comunidad estratigráfica está considerando la formalización del Antropoceno como una nueva época geológica, pero independientemente del resultado de ese proceso, se está volviendo evidente que las condiciones del Antropoceno transgreden las condiciones del Holoceno en varios aspectos. El conocimiento de que la actividad humana ahora rivaliza con las fuerzas geológicas en la influencia de la trayectoria del Sistema Tierra tiene implicaciones importantes tanto para la ciencia del Sistema Tierra como para la toma de decisiones sociales.



La Secretaría del Trabajo y Previsión Social deberá pronunciarse respecto de la solicitud de registro dentro de los veinte días posteriores a la recepción de la misma, de no hacerlo, los solicitantes podrán requerirla para que dicte la resolución correspondiente, dentro de los tres días siguientes a la presentación del requerimiento. Transcurrido dicho plazo sin que se notifique la resolución, se tendrá por efectuado el registro para los efectos legales a que dé lugar. [Artículo 15. LEY FEDERAL DEL TRABAJO, publicada 30-09-2024]

Google Translator - Legal Text



The Ministry of Labor and Social Welfare (MLSW) must approve or reject the registration request within twenty days after its receipt in the offices of the MLSW. If the MLSW fails to do so, the applicants may formally require that the MLSW resolve the issue within three business days. If, after this second period has elapsed without notification of the status of the registration request, the registration request will be deemed approved for all applicable legal purposes. [*Marron translation*]

ChatGPT - Legal Text

ChatGPT ▾

⬆ Share

Translate the following text from Spanish to English as a professional translator

< 2/2 >



Please provide the text you'd like me to translate, and I'll deliver a professional-quality translation for you!



The Ministry of Labor and Social Welfare must issue a decision regarding the registration request within twenty days of receiving it. If it fails to do so, the applicants may formally request that the corresponding resolution be issued within three days following the submission of such a request. If this period elapses without notification of the resolution, the registration will be considered completed for all applicable legal purposes.

Registered User AI Translators (pay per limit)

OpenAI API: GPT-4o

GPT-4o: A type of foundational LLM that can generate content in over 90 languages. Multiple types of rate limits apply.

“We’re running GPT4 translations on english, german, french, spanish, italian, portugese, on a daily basis, and I can assure you it’s the best available option hands down.”

“...the talk of the town with its very impressive language translation technology that’s taking multilingual communication by storm. GPT-4 has the remarkable ability to analyze the context and meaning of words and sentences, which makes it able to make more accurate translations”

“...it generates text 2x faster and is 50% cheaper. Additionally, GPT-4o has the best vision and performance across non-English languages of any of our models. **GPT-4o is available in the OpenAI API to paying customers.**”

Access to GPT-4o API

Once you have applied for access, you must be reviewed and approved...

Once your application has been approved, you will receive an API key ...

Once you have gained access to the API, you will need to install the necessary SDK (using an IDE)

Once you have configured GPT-4 for your project, you need to initialize GPT-4 for translation....

And now you will be ready to start using the model for language translation.

Yikes!

Google Cloud Translation API

This API uses a pre-trained, custom model or a translation specialized large language model (LLMs). You can choose from different models based on your needs, such as:

Neural Machine Translation (NMT): For general text, like website content or news articles. Available in 194 languages!

Translation Large Language Model (LLM): For conversational text, like messages or social media posts

AutoML Translation: For highly specialized or technical content

NB. Google Translate uses sequence-to-sequence models. These models also are used in voice-activated gadgets, and online chatbots.

AI vs. Human Translators: The Science

GPT-4 vs. Human Translators: A Comprehensive Evaluation of Translation Quality Across Languages, Domains, and Expertise Levels

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³ University College London ⁴ University of Cambridge ⁵ Lan-Bridge Group

⁶ Institute of Advanced Technology, Westlake Institute for Advanced Study

To our knowledge, **this study is the first to evaluate LLMs against human translators** and analyze the systematic differences between their outputs, providing valuable insights into the current state of LLM-based translation and its potential limitations.

GPT-4 vs. Human Translators: A Comprehensive Evaluation of Translation Quality Across Languages, Domains, and Expertise Levels

ABSTRACT This study comprehensively evaluates the translation quality of Large Language Models (LLMs), specifically GPT-4, against human translators of varying expertise levels across multiple language pairs and domains... we find that **GPT-4 performs comparably to junior translators in terms of total errors made but lags behind medium and senior translators....**[we] find that **GPT-4 translator suffers from literal translations, but human translators sometimes overthink the background information.**

GPT-4 vs. Human Translators: A Comprehensive Evaluation of Translation Quality Across Languages, Domains, and Expertise Levels

CONCLUSION ... GPT-4 has made significant strides in approaching human-level translation quality...This suggests promising opportunities for collaboration and enhancement of translation workflows...we anticipate that **LLMs will become increasingly valuable tools in the translation industry, working alongside human translators to improve productivity, efficiency, and overall translation quality.**

Natural Language Translation in the 21st C.

Is AI a Threat to Human Language Translators?

Perhaps, but unlikely unless folks are unprepared.

-
What is likely is that professional translators will increasingly be required to interface with LLMs through subscription-based APIs. This means adding machine language coding to your tool kit.

Using Google Cloud Translate API

Not trivial!!

My Adventures with Google Translate API: google-cloud-sdk and google-cloud-translate

The screenshot shows the Google Cloud console interface. At the top, the browser address bar displays the URL `https://console.cloud.google.com/welcome/new?pli=1&project=my-project-utec1`. Below the address bar, a navigation bar includes links to Gmail, 4G CPE, OnPoint, Bank of America, SSA, SAT, Mi Telcel, WhatsApp, Google_Cloud, UTECA_MSOoffice, and UTECA_Teams. A search bar is also present. The main content area features a welcome message: "Welcome, Bruce Marron". To the left of the message is a green circular progress indicator with a shopping bag icon, indicating the trial status. To the right is a blue box with information about the project and a link to try the Gemini 1.5 Pro model. The trial status box shows that \$24 out of \$6,220 credits have been used, with the trial expiring on March 24, 2025. It includes links to "What happens when trial ends?", "Add people to your project", "Set up budget alerts", and "Review product spend". The project information box shows the project name "My Project-UTECA1", the number "735387290281", and the ID "my-project-utec1". It also includes a link to "Try our most advanced model: Gemini 1.5 Pro" and a button to "Try Gemini".

← → ↺ 🏠 `https://console.cloud.google.com/welcome/new?pli=1&project=my-project-utec1` ☆ 📧 📄 👤 📁 ≡

Gmail 4G CPE OnPoint Bank of America SSA SAT Mi Telcel WhatsApp Google_Cloud UTECA_MSOoffice UTECA_Teams >> 📁 Other Bookmarks

📦 Free trial status: \$6,195.58 credit and 74 days remaining. Activate your full account to get unlimited access to all of Google Cloud—use any remaining credits, then pay only for what you use. **DISMISS** **ACTIVATE**

≡ Google Cloud My Project-UTECA1 Search (/) for resources, docs, products, and more 🔍 Search ✦ 📄 🔔 ? 🌐

 Welcome, Bruce Marron

You're in Free Trial

 \$24 out of \$6,220 credits used
Expires March 24, 2025
[What happens when trial ends?](#)
Activate full account

You're working on project [My Project-UTECA1](#) ⓘ
Number: 735387290281 📄 ID: my-project-utec1 📄
[Add people to your project](#)
[Set up budget alerts](#)
[Review product spend](#)

Try our most advanced model:
Gemini 1.5 Pro

Try Gemini →

My Adventures with Google Translate API: google-cloud-sdk and google-cloud-translate

The image shows a code editor window with the following content:

```
58 '''  
59 References for  
60 Google Cloud Translation API with Python  
61 '''  
62 # Google Account  
63 https://cloud.google.com/iam/docs/service-accounts-create  
64  
65 # Install Google Cloud SDK (or CLI)  
66 https://cloud.google.com/sdk/docs/initializing  
67 https://stackoverflow.com/questions/71086225/how-to-install-or-uninstall-gcloud  
68  
69 # Service Account (not used)  
70 https://cloud.google.com/iam/docs/service-account-overview  
71  
72 # Google Authentication and Application Default Credentials (ADC)  
73 https://cloud.google.com/docs/authentication  
74 https://cloud.google.com/docs/authentication/set-up-adc-local-dev-environment  
75 https://cloud.google.com/docs/authentication/application-default-credentials#pers  
76 https://cloud.google.com/docs/authentication/set-up-adc-local-dev-environment  
77 https://cloud.google.com/docs/authentication/provide-credentials-adc  
78 https://cloud.google.com/docs/authentication/provide-credentials-adc#how-to  
79 https://cloud.google.com/docs/authentication/application-default-credentials  
80 https://stackoverflow.com/questions/51554341/google-auth-exceptions-default creden  
81 https://stackoverflow.com/questions/40932678/where-are-google-application-default  
82 https://medium.com/datamindedbe/application-default-credentials-477879e31cb5  
83
```

The right side of the image shows a Jupyter Notebook interface with the following content:

home/bmarron

No variables to show

Run code in the Editor or IPython console to see any global variables listed here for exploration and editing.

Help Variable Explorer Debugger Plots Files

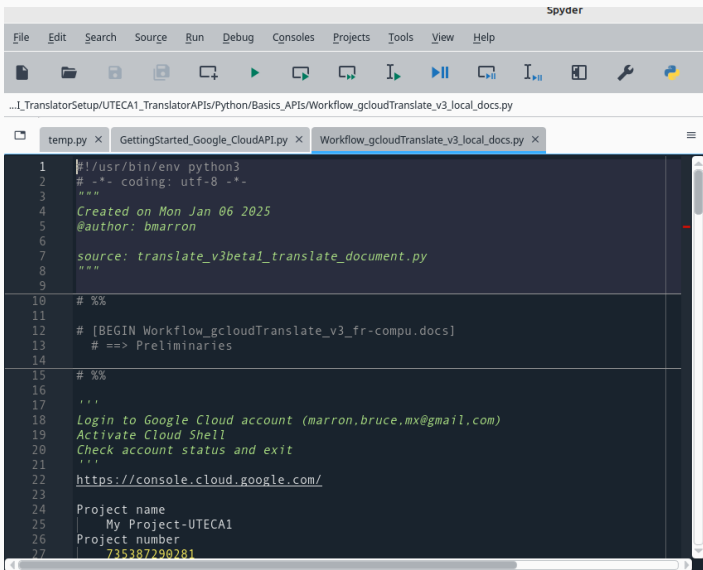
Console 1/A x

Type "copyright", "credits" or "license" for more information.

IPython 8.30.0 -- An enhanced Interactive Python
Type '?' for help.

In [1]:

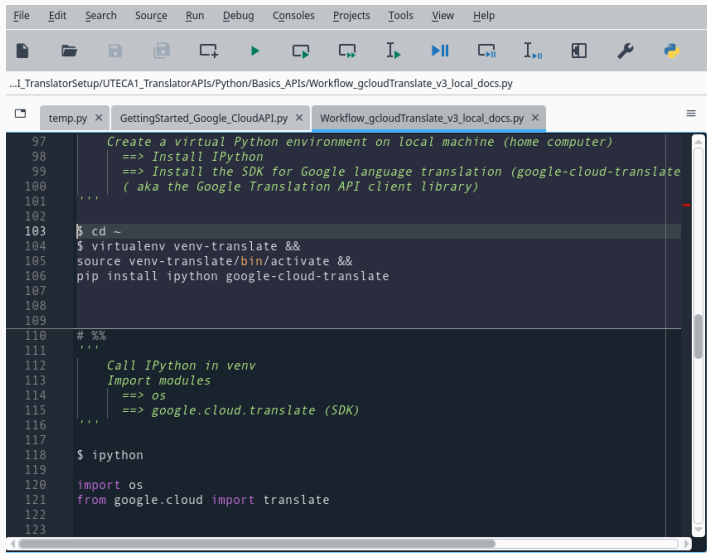
My Adventures with Google Translate API: google-cloud-sdk and google-cloud-translate



The image shows the Spyder IDE interface. The top menu bar includes File, Edit, Search, Source, Run, Debug, Consoles, Projects, Tools, View, and Help. Below the menu is a toolbar with icons for file operations and execution. The main editor window displays a Python script with the following content:

```
1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Created on Mon Jan 06 2025
5  @author: bmarron
6
7  source: translate_v3beta1_translate_document.py
8  """
9
10 # %%
11
12 # [BEGIN Workflow_gcloudTranslate_v3_fr-compu.docs]
13 # ==> Preliminaries
14
15 # %%
16
17 '''
18 Login to Google Cloud account (marron.bruce,mx@gmail.com)
19 Activate Cloud Shell
20 Check account status and exit
21 '''
22 https://console.cloud.google.com/
23
24 Project name
25 |   My Project-UTECA1
26 Project number
27 |   735387290281
```

My Adventures with Google Translate API: google-cloud-sdk and google-cloud-translate

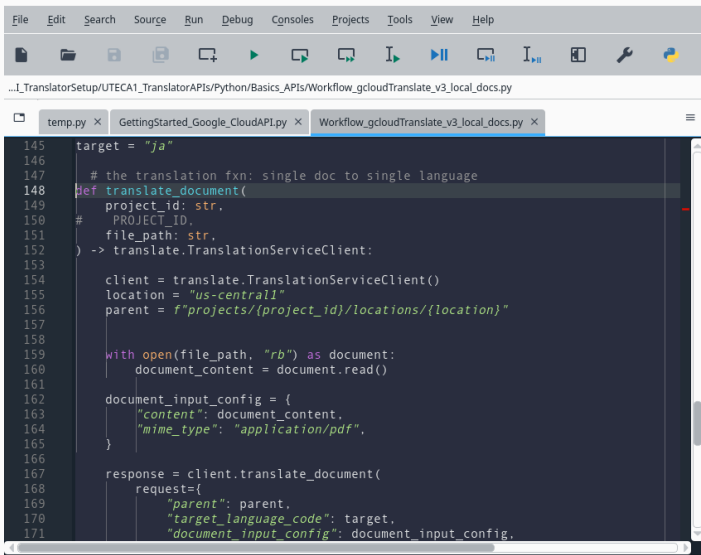


```
File Edit Search Source Run Debug Consoles Projects Tools View Help

temp.py x GettingStarted_Google_CloudAPI.py x Workflow_gcloudTranslate_v3_local_docs.py

103 Create a virtual Python environment on local machine (home computer)
104 ==> Install IPython
105 ==> Install the SDK for Google language translation (google-cloud-translate
106 ( aka the Google Translation API client library)
107 ...
108 $ cd ~
109 $ virtualenv venv-translate &&
110 source venv-translate/bin/activate &&
111 pip install ipython google-cloud-translate
112
113 # %%
114 ...
115 Call IPython in venv
116 Import modules
117 ==> os
118 ==> google.cloud.translate (SDK)
119 ...
120 $ ipython
121 import os
122 from google.cloud import translate
123
```

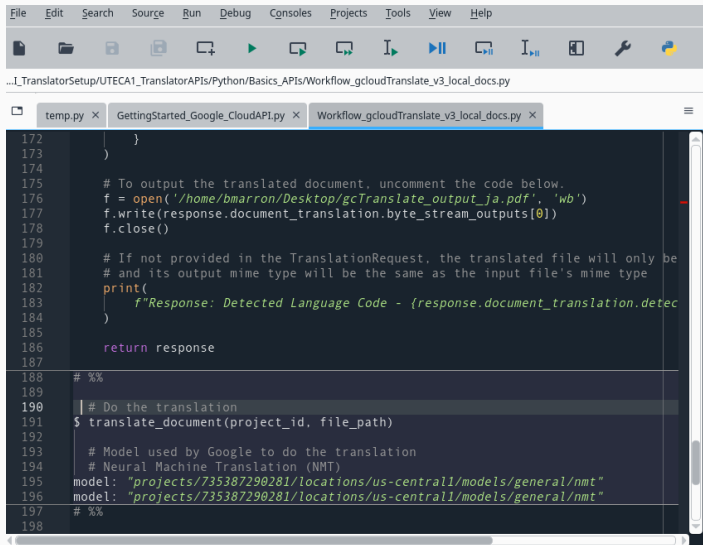

My Adventures with Google Translate API: google-cloud-sdk and google-cloud-translate



The screenshot shows an IDE window with the file path `...I_TranslatorSetup/UTECA1_TranslatorAPIs/Python/Basic_APIs/Workflow_gcloudTranslate_v3_local_docs.py`. The code is written in Python and defines a `translate_document` function. The function takes `project_id`, `file_path`, and `target` as arguments. It uses the `google-cloud-translate` library to create a `TranslationServiceClient` and translate the content of the file at `file_path` into the language specified by `target`. The code is as follows:

```
145 target = "ja"
146
147 # the translation fxn: single doc to single language
148 def translate_document(
149     project_id: str,
150     # PROJECT_ID.
151     file_path: str,
152 ) -> translate.TranslationServiceClient:
153
154     client = translate.TranslationServiceClient()
155     location = "us-central1"
156     parent = f"projects/{project_id}/locations/{location}"
157
158     with open(file_path, "rb") as document:
159         document_content = document.read()
160
161     document_input_config = {
162         "content": document_content,
163         "mime_type": "application/pdf",
164     }
165
166     response = client.translate_document(
167         request={
168             "parent": parent,
169             "target_language_code": target,
170             "document_input_config": document_input_config,
```

My Adventures with Google Translate API: google-cloud-sdk and google-cloud-translate



The screenshot shows a code editor window with three tabs: 'temp.py', 'GettingStarted_Google_CloudAPI.py', and 'Workflow_gcloudTranslate_v3_local_docs.py'. The active tab is 'Workflow_gcloudTranslate_v3_local_docs.py'. The code is written in Python and includes comments in Spanish. It defines a function to translate a document using the Google Cloud Translate API. The code includes a comment about outputting the translated document to a PDF file and another comment about the default model used for translation. The code is as follows:

```
172     }
173 }
174
175 # To output the translated document, uncomment the code below.
176 f = open('/home/bmarron/Desktop/gcTranslate_output_ja.pdf', 'wb')
177 f.write(response.document_translation.byte_stream_outputs[0])
178 f.close()
179
180 # If not provided in the TranslationRequest, the translated file will only be
181 # and its output mime type will be the same as the input file's mime type
182 print(
183     f"Response: Detected Language Code - {response.document_translation.detec
184 )
185
186 return response
187
188 # %%
189
190 | # Do the translation
191 $ translate_document(project_id, file_path)
192
193 # Model used by Google to do the translation
194 # Neural Machine Translation (NMT)
195 model: "projects/735387290281/locations/us-central1/models/general/nmt"
196 model: "projects/735387290281/locations/us-central1/models/general/nmt"
197 # %%
198
```

My Adventures with Google Translate API: google-cloud-sdk and google-cloud-translate



Marrón
Spring 2025

Course Title: **Taller de Traducción Ingeniería y Adelantos Tecnológicos**
Program and Track: **Interpretación y Traducción: 6° Cuatrimestre**
Course ID: **IT0629**

Description

This course provides the modern foundations for linguistic translation in the engineering sciences and in the broader field of advanced technologies. The course will focus on translation in selected engineering disciplines as well as translation related to advanced and emerging technologies that come primarily from the engineering sciences. The professional translational mapping tools developed in this course will be specific to Spanish and English translations, although such tools and methodologies can readily be extended to the translation of other languages.

Major elements of the course will include: i) the conceptual framework for the science of engineering; ii) the structure, function, and dominant paradigm of selected engineering disciplines; iii) the state of advanced and emerging technologies in selected engineering disciplines; iv) the elements of style and the use of standard conventions in English; v) a review of essential English grammatical structures; vi) Version Control Systems (VCSs) and elements of quality assurance and quality control (QA/QC) applied to documents; vii) the professional use of Large Language Models (LLMs) in translation; viii) the direct translation of engineering documents; and ix) the production of client deliverables.

Scope

1. Conceptual Framework for the Science of Engineering
 - 1.1 Historical Context
 - 1.2 Science vs. Engineering
 - 1.3 Major Engineering Disciplines
2. Structure, Function, and Dominant Paradigms
 - 2.1 Thomas Kuhn's "The Structure of Scientific Revolutions"
 - 2.2 Analysis of Selected Engineering Disciplines
3. The Elements of Style in Professional Writing
 - 3.1 Strunk and White's, "Elements of Style"
 - 3.2 Developing a Professional Voice
4. Review of Essential English Grammar
 - 4.1 Murphy's, "English Grammar in Use"
 - 4.2 The Yale Graduate Writing Center
5. Document QA/QC and Version Control Systems
 - 5.1 Introduction to Document QA/QC
 - 5.1.1 Fundamental Concepts of QA/QC
 - 5.1.2 Frameworks for Document QA/QC
 - 5.2 Introduction to Version Control Systems (VCSs)
 - 5.2.1 Using Git and Github

Course Description in .pdf format
Original language: English

My Adventures with Google Translate API: google-cloud-sdk and google-cloud-translate

Machine Translated by Google



Marron
Printemps 2025

Titre du cours : Abiliter de traduction l'ingénierie et avancées technologiques
Programme et parcours : Interprétation et traduction : 6e semestre
ID du cours : IT0629

Description

Ce cours fournit les bases modernes de la traduction linguistique dans les sciences de l'ingénierie et dans le domaine plus large des technologies avancées. Le cours se concentrera sur la traduction dans certaines disciplines d'ingénierie ainsi que sur la traduction liée aux technologies avancées et émergentes qui proviennent principalement des sciences de l'ingénierie. Les outils professionnels de cartographie translationnelle développés dans ce cours. Le cours sera spécifique aux traductions en espagnol et en anglais, bien que ces outils et méthodologies puissent facilement être étendus à la traduction d'autres langues.

Les principaux éléments du cours comprendront : i) le cadre conceptuel de la science de l'ingénierie ; ii) la structure, la fonction et le paradigme dominant de certaines disciplines d'ingénierie ; iii) l'état des technologies avancées et émergentes dans certaines disciplines d'ingénierie ; iv) les éléments de style et l'utilisation des conventions standard en anglais ; v) un examen des structures grammaticales anglaises essentielles. vi) les systèmes de contrôle de version (VCS) et les éléments d'assurance qualité et de contrôle qualité (QAQC) appliqués aux documents ; vii) l'utilisation professionnelle de grands modèles linguistiques (LLM) dans la traduction ; viii) la traduction directe de documents d'ingénierie ; et ix) la production de livrables clients.

Préface

1. Cadre conceptuel de la science de l'ingénierie
 - 1.1 Contexte historique
 - 1.2 Science et ingénierie
 - 1.3 Principales disciplines d'ingénierie
2. Structure, fonction et paradigmes dominants
 - 2.1 « La structure des révolutions scientifiques » de Thomas Kuhn
 - 2.2 Analyse de certaines disciplines d'ingénierie
3. Les éléments de style dans la rédaction professionnelle
 - 3.1 Strunk et White. « Éléments de style »
 - 3.2 Développer une voix professionnelle
4. Révision de la grammaire anglaise essentielle
 - 4.1 Murphy. « La grammaire anglaise en usage »
 - 4.2 Le Centre d'écriture des diplômés de Yale
5. Document QAQC et systèmes de contrôle de version
 - 5.1 Introduction au document QAQC
 - 5.1.1 Concepts fondamentaux de l'AQC
 - 5.1.2 Cadres pour l'AQC des documents
 - 5.2 Introduction aux systèmes de contrôle de version (VCS)
 - 5.2.1 Utilisation de Git et GitHub

Course Description in .pdf format
Translated language: French

My Adventures with Google Translate API: google-cloud-sdk and google-cloud-translate

Machine Translated by Google



Braun
Frühling 2025

Kursitel: Übersetzungswerkstatt Technik und technologische Fortschritte
Programm und Track: Dolmetschen und Übersetzen: 6. Semester
Kurs-ID: IT0629

Beschreibung

Dieser Kurs vermittelt die modernen Grundlagen für sprachliche Übersetzungen in den Ingenieurwissenschaften und im weiteren Bereich der fortschrittlichen Technologien. Der Kurs konzentriert sich auf Übersetzungen in ausgewählten Ingenieurdisziplinen sowie auf Übersetzungen im Zusammenhang mit fortschrittlichen und aufkommenden Technologien, die hauptsächlich aus den Ingenieurwissenschaften stammen. Die in diesem Kurs entwickelten professionellen Translationsmapping-Tools. Der Kurs konzentriert sich speziell auf Übersetzungen ins Spanische und Englische, obwohl diese Werkzeuge und Methoden problemlos auf die Übersetzung anderer Sprachen ausgeweitet werden können.

Wichtige Elemente des Kurses sind: i) der konzeptionelle Rahmen der Ingenieurwissenschaften; ii) Struktur, Funktion und vorherrschendes Paradigma ausgewählter Ingenieurdisziplinen; iii) der Stand fortschrittlicher und aufkommender Technologien in ausgewählten Ingenieurdisziplinen; iv) Stilelemente und Verwendung standardmäßiger Konventionen in der englischen Sprache; v) eine Wiederholung wesentlicher englischer Grammatikstrukturen; vi) Auf Dokumente angewendete Versionskontrollsysteme (VCSs) und Elemente der Qualitätssicherung und Qualitätskontrolle (QA/QC); vii) die professionelle Verwendung von Large Language Models (LLMs) bei der Übersetzung; viii) die direkte Übersetzung von technischen Dokumenten; und ix) die Erstellung von Liefergegenständen für den Kunden.

Umfang

1. Konzeptioneller Rahmen für die Ingenieurwissenschaften
 - 1.1 Historischer Kontext
 - 1.2 Wissenschaft vs. Technik
 - 1.3 Wichtige Ingenieurdisziplinen
2. Struktur, Funktion und dominante Paradigmen
 - 2.1 Thomas Kuhns „Die Struktur wissenschaftlicher Revolutionen“
 - 2.2 Analyse ausgewählter Ingenieurdisziplinen
3. Die Elemente des Stils beim professionellen Schreiben
 - 3.1 Shunk und White, „Elements of Style“
 - 3.2 Entwicklung einer professionellen Stimme
4. Wiederholung der grundlegenden englischen Grammatik
 - 4.1 Murphy's, „Englische Grammatik im Einsatz“
 - 4.2 Das Yale Graduate Writing Center
5. Dokumentieren Sie QA/QC und Versionskontrollsysteme
 - 5.1 Einführung zum Dokument QA/QC
 - 5.1.1 Grundlegende Konzepte der Qualitätssicherung und Qualitätskontrolle
 - 5.1.2 Rahmenbedingungen für die Dokumenten Qualitätsicherung Qualitätskontrolle
 - 5.2 Einführung in Versionskontrollsysteme (VCSs)
 - 5.2.1 Verwendung von Git und GitHub

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茶色
2025年春

コース名: 翻訳ワークショップ エンジニアリングと技術の進歩
プログラムとトラック: 通訳翻訳 16学期
コースID: IT0629

説明

このコースでは、工学科学とより広範な先端技術分野における言語翻訳の現代的な基礎を提供します。このコースでは、特定の工学分野の翻訳と、主に工学科学から生まれた先端技術や新興技術に関する翻訳に焦点を当てます。このコースで関与された専門的な翻訳マッピングツールは、

このコースはスペイン語と英語の翻訳に特化していますが、このようなツールと方法は他の言語の翻訳にも容易に拡張できます。

コースの主な要素には、i) 工学の概念の枠組み、ii) 選択された工学分野の構造、機能、および支配的なパラダイム、iii) 選択された工学分野における先端技術と新興技術の状態、iv) 英語のスタイル要素と標準規則の使用、v) 基本的な英語の文法構造のレビューが含まれます。

vi) 文書に適用されるバージョン管理システム (VCS) と品質保証および品質管理 (QA/QC) の要素、vii) 翻訳における大規模言語モデル (LLM) の専門的な使用、viii) エンジニアリング文書の直訳翻訳、および ix) クライアント産業物の作成。

範囲

1. 工学科学の概念的枠組み
 - 1.1 歴史的背景
 - 1.2 科学と工学
 - 1.3 主な工学分野
2. 構造、機能、そして支配的なパラダイム
 - 2.1 トーマス・エディンガムの「科学革命の構造」
 - 2.2 選択された工学分野の分析
3. プロフェッショナルライティングにおけるスタイルの要素
 - 3.1 ストラクチャーとホワイトの「スタイルの要素」
 - 3.2 プロフェッショナルな声を育てる
4. 基本的な英語文法の復習
 - 4.1 マーフィーの「English Grammar in Use」
 - 4.2 イェール大学大学院ライティングセンター
5. ドキュメントQA/QCとバージョン管理システム
 - 5.1 ドキュメントQA/QCの概要
 - 5.1.1 QA/QCの基本概念
 - 5.1.2 ドキュメントQA/QCのフレームワーク
 - 5.2 バージョン管理システム (VCS)の概要
 - 5.2.1 GitとGithubの使用

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भूत
साल 2025

पाठ्यक्रम कीर्ति:
कार्यक्रम और टैग:
पाठ्यक्रम अंश:

अनुवाद इंजीनियरिंग और जलवायु प्रणालि पर कार्यवाही
आकाश और अनुवाद: सटी विषय
अंश: 2025

विषय

यह पाठ्यक्रम इंजीनियरिंग विज्ञान और उच्च प्रौद्योगिकियों के माध्यम से पाठ्य अनुवाद के लिए आधुनिक अवधारणा प्रदान करता है। यह पाठ्यक्रम कर्मीय इंजीनियरिंग विषयों में अनुवाद के साथ-साथ उच्च और उपरती प्रौद्योगिकियों से संबंधित अनुवाद पर ध्यान केंद्रित करता है। मुख्य रूप से इंजीनियरिंग विज्ञान से आते हैं। इस पाठ्यक्रम में निम्नलिखित क्षेत्रों में अनुवाद शामिल है: अनुवाद प्रणालियों का उपयोग।

पाठ्यक्रम केवल संश्लेष और अंशों के अनुवादों तक ही सीमित होना, हालांकि ऐसे उपकरणों और पद्धतियों को अन्य भाषाओं के अनुवाद के लिए भी आसानी से लागू किया जा सकता है।

पाठ्यक्रम के प्रमुख तत्वों में शामिल हैं: i) इंजीनियरिंग विज्ञान के लिए वैश्विक दृष्टिकोण; ii) कर्मीय इंजीनियरिंग विषयों की संरचना, कार्य और प्रमुख प्रणालियाँ; iii) कर्मीय इंजीनियरिंग विषयों में उच्च और उपरती प्रौद्योगिकियों की प्रणालि; iv) शैली के उच्च और अंशों में पाठ्य प्रणालियों का उपयोग; v) आवश्यक अंशों के व्यावहारिक संरचनाओं की समीक्षा।

vi) दस्तावेजों पर लागू संरचना प्रणालि प्रणालि (सीसीएन) और पुनर्रचना अनुवाद (अनुवादप्रणालि) के उच्च; vii) अनुवाद में बड़े भाषा मॉडल (एलएलएम) का व्यावहारिक उपयोग; viii) इंजीनियरिंग दस्तावेजों का शैली अनुवाद; और ix) दृष्टिकोण विषय का उपयोग।

प्रणालि

1. इंजीनियरिंग विज्ञान के लिए वैश्विक दृष्टिकोण
 - 1.1 वैश्विक दृष्टिकोण
 - 1.2 विज्ञान का उपयोग इंजीनियरिंग
 - 1.3 प्रमुख इंजीनियरिंग विषय
2. संरचना, कार्य और प्रमुख प्रणालि
 - 2.1 संरचना का उपयोग इंजीनियरिंग
 - 2.2 कर्मीय इंजीनियरिंग विषयों का उपयोग
3. व्यावहारिक प्रणालि प्रणालि में शैली के उच्च
 - 3.1 दृष्टिकोण और प्रणालि: "शैली के उच्च"
 - 3.2 व्यावहारिक प्रणालि प्रणालि का उपयोग
4. आवश्यक अंशों के व्यावहारिक की समीक्षा
 - 4.1 दृष्टिकोण, "इंजीनियरिंग का उपयोग"
 - 4.2 दृष्टिकोण का उपयोग
5. QA/QC और संरचना प्रणालि प्रणालि का दस्तावेजीकरण करें
 - 5.1 दस्तावेजी QA/QC का उपयोग
 - 5.1.1 QA/QC की प्रणालि प्रणालि
 - 5.1.2 दस्तावेजी QA/QC के लिए उपयोग
 - 5.2 दस्तावेजी प्रणालि (VCS) का उपयोग
 - 5.2.1 Git और GitHub का उपयोग

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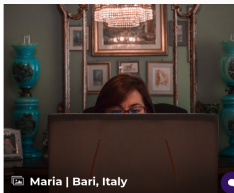


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